Data Structures & Algorithms Lab: Solved FAQs, Experiments 6 to 9

SY B.Tech E&TC, Semester I, 2026-27

Experiment 8 has no FAQ section in the manual, and Experiment 9's FAQs cover trees (the manual's Exp 9 content does not match its Prim's title).

Experiment 6: Queue (and Stack with Linked List)

1. **What is a queue?** A linear data structure (FIFO) where insertion happens at the rear and deletion at the front.
2. **How is a queue represented?** Using an array or a linked list.
3. **Types of queues?** Linear queue, circular queue, priority queue, deque.
4. **Concept of circular queue?** The last position is connected back to the first, so the rear wraps around (`rear = (rear+1) % MAX`). This reuses freed space.
5. **Applications of queues?** CPU and disk scheduling, printer spooling, buffering (such as the UART buffer in Exp 10), BFS, queue simulation.
6. **What is a priority queue?** Each element has a priority. The higher-priority element is processed first; elements with equal priority are processed in the order they arrived.
7. **What is a deque?** A double-ended queue. Elements can be inserted or removed at both the front and the rear, but not in the middle.

Experiment 7: Postfix Evaluation

1. **Types of expressions?** Infix, prefix and postfix.
2. **Examples of each?** Infix: `a+b`. Prefix: `+ab`. Postfix: `ab+`.
3. **Data structure used for evaluation?** Stack.
4. **Infix a+b+c+d to prefix?** `+++abcd`
5. **Infix a+b+c+d to postfix?** `ab+c+d+`

Experiment 8: Binary Search Tree (Inorder Traversal)

No FAQs are given in the manual for this experiment.

Experiment 9: Trees (as given in the manual)

1. **What is a tree?** A non-linear data structure that stores hierarchical data as nodes connected by edges, with one root node.
2. **What is a binary tree?** A tree that is empty, or consists of a root and two disjoint binary trees called the left and right subtrees. Each node has at most two children.
3. **Types of binary trees?** Full, complete, skewed, binary search tree, threaded, AVL and expression trees.
4. **Tree traversal techniques?** Recursive and non-recursive (iterative).
5. **Recursive traversal techniques?** Inorder (Left, Root, Right), Preorder (Root, Left, Right), Postorder (Left, Right, Root).
6. **What are AVL trees?** Self-balancing binary search trees in which the height difference between the left and right subtrees of every node is at most 1.
7. **Basic operations on a tree?** Create (insert), search, delete, traversal.
8. **Degree and height of a tree?** The degree of a node is its number of subtrees (children). The height or depth of a tree is the maximum level of any node.
9. **Representation of a tree?**
   - **Linked list:** each node is `struct node { int data; struct node *left, *right; };`
   - **Array:** root at index 1. For a node at index `i`, left child is at `2i`, right child at `2i+1`, parent at `i/2`. Missing children leave empty slots.